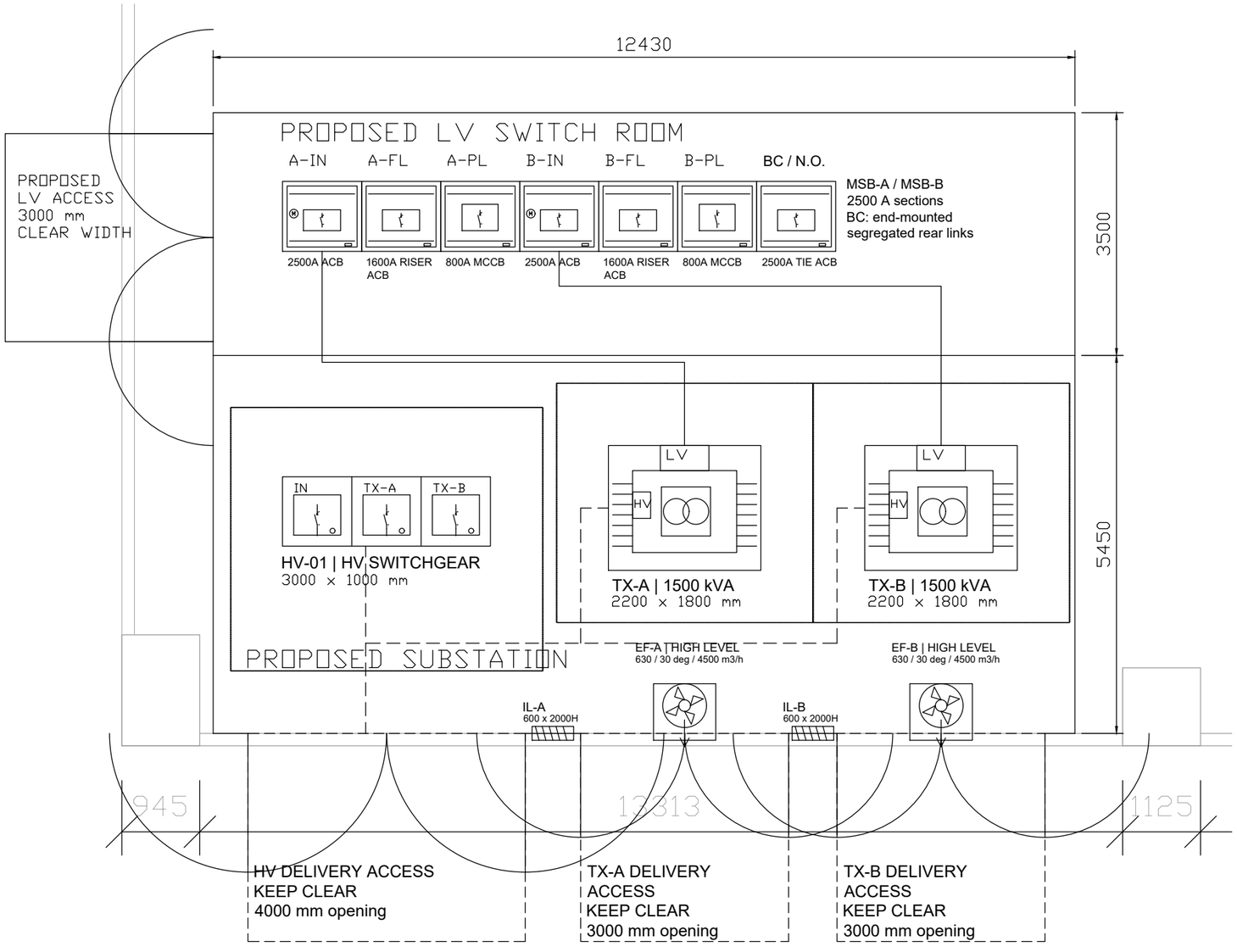
TASK 1 | PROPOSED SUBSTATION AND MAIN LV ROOM

A3 LANDSCAPE | SCALE 1:75 | DIMENSIONS IN mm | PLANTROOM DETAIL



LINE KEY: dashed line = HV trench; solid line = LV route; dashed box = delivery clearance. Equipment details diagrammatic.

LV INTERFACES / HV ROUTES

LV trench: 1000 W x 1100D, four MCTs per interface. A: 8.03+2=10.03m; B: 8.31+2=10.31m (<20m). 2m is assumed total termination/vertical allowance.

HV centre-lines: 1000 W at panels; 600 W branches; final 300 W; generally 1400D. Segregate HV/LV. Covered trenches; no beam obstruction.

DETAIL 1:75 AT A3

EQUIPMENT

TX-A/B: CLP 1500 kVA each; demand 2672.216 kVA.

HV-01: incomer + two transformer protection panels.

MSB: two 2500 A sections, seven 1140 mm bays.

BC: normally-open 4P end coupler with segregated rear links. See main schematic for interlocking.

CLEARANCES / BUILDING WORKS

HV: rear 1000, front 1800, sides 750 mm.

TX: LV terminals 900, other sides 750 mm.

LV board: front 1500, rear 990 mm; total width 7980 mm.

Proposed 2 h FRR partitions, sealed penetrations and FFL 150 mm above external level.

ACCESS AND DOORS

Proposed south delivery access: >=3000 mm clear route; unloading within 20 m.

HV opening 4000 mm; TX openings 3000 mm each; proposed height 2800 mm.

Outward-opening doors shown at 90 degrees; adjacent doors used sequentially.

TX-A wicket access; independent west LV entrance.

VENTILATION

EF-A/B: 630 mm / 30 deg / 700 rpm; each 4500 m3/h at 70 Pa; <=58 dB(A) at 1 m (COP101 6.6).

IL-A/B: 600 x 2000H mm, >=50% free area; total 1.20 m2.

High-level ducts: 900 x 750 mm; discharge >=2.5 m above street. Thermal calculation: Task 1.

CLP COP215 Rev07; COP101 v16, 4.3 and 6.1-6.7.

Equipment selections and calculations accompany this drawing.